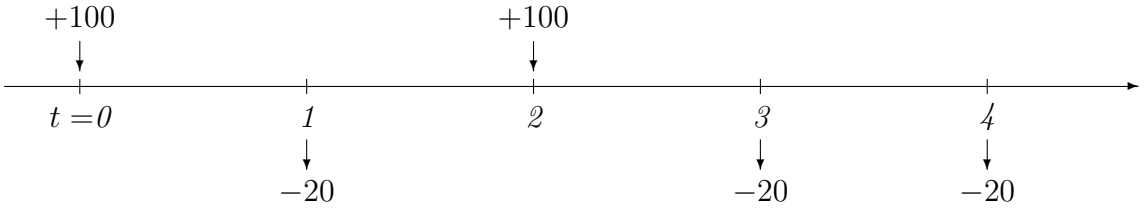
F78AA - Actuarial and Financial Mathematics A

Tutorial 2 – Valuing cashflows and annuities

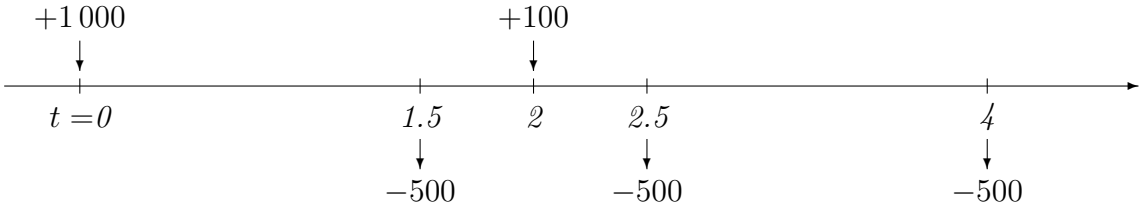
Question 1 The effective rate of interest is 6% per time unit. Cashflows are as follows (the time axis is calibrated in time units):



For these cashflows, calculate:

1. the accumulation at time $t = 5$ time units.
2. the accumulation at time $t = 10$ time units.
3. the present value at time $t = 0$ time units.

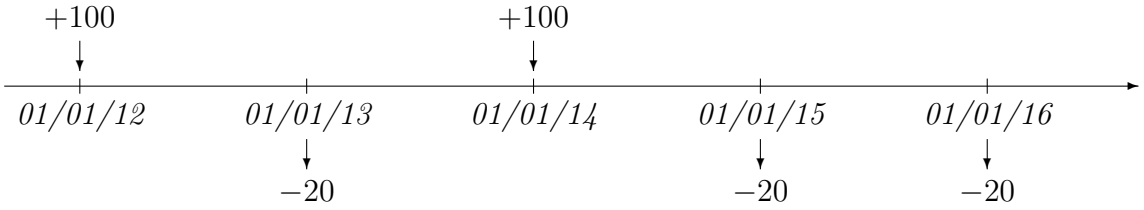
Question 2 The effective rate of interest is 8% per annum. Cashflows are as follows (the time axis is calibrated in years):



For these cashflows, calculate:

1. the total accumulation at time $t = 5$ years.
2. the total present value at time $t = 0$ years.
3. the total value at time $t = 2$ years.

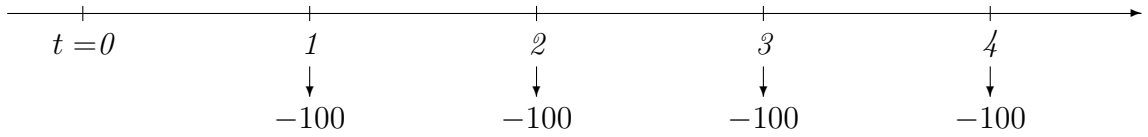
Question 3 The effective rate of interest is 3% per annum during 2012 and 2013, 2% per half-year until 1 July 2015, and 1% per month thereafter. Cashflows are as follows (the time axis is calibrated in calendar time):



For these cashflows, calculate the total:

1. present value on 01/01/12.
2. accumulation on 01/05/17.
3. value on 01/07/15.

Question 4 The effective rate of interest is 5.5% per time unit. You have a liability to pay the cashflows as shown in the following diagram (the time axis is calibrated in time units) at the end of the 1st, 2nd, 3rd and 4th time units:



In order to discharge this liability, you set up a bank account time $t = 0$ and pay into it an amount £X. You will meet the future payments out of this bank account.

Calculate the value of X that will allow you to exactly meet all the future payments.

Question 5

1. Describe in words the meaning of the following symbols:

   (a) $s^{i}_{\overline{10|}}$ (b) $\ddot{s}^{i}_{\overline{25|}}$ (c) $a^{i}_{\overline{10|}}$ (d) $\ddot{a}^{i}_{\overline{20|}}$

2. Assuming $i = 0.07$ per time unit, calculate the values of the above symbols.

Question 6 Prove the following:

1. $s_{\overline{n|}} = (1+i)^n a_{\overline{n|}}$.
2. $s_{\overline{n|}} = 1 + \ddot{s}_{\overline{n-1|}}$.
3. $\ddot{a}_{\overline{n|}} = v^n \ddot{s}_{\overline{n|}}$.
4. $\ddot{a}_{\overline{n|}} = 1 + a_{\overline{n-1|}}$.

Solutions: Q1.1. 184.00; Q1.2. 246.24; Q1.3. 137.50; Q2.1 -205.35; Q2.2 -139.75; Q2.3 -163.01; Q3.1 139.99; Q3.2 196.17; Q3.3 157.60; Q4.1 350.51; Q5.2 (a) 13.81645; Q5.2 (b) 67.67647; Q5.2 (c) 7.02358; Q5.2 (d) 11.33560.

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Solutions to Tutorial 2 – Valuing cashflows and annuities

Solution 1

1. The accumulation to $t = 5$ is:

$$\begin{aligned} & 100V(5,0) - 20V(5,1) + 100V(5,2) - 20V(5,3) - 20V(5,4) \\ = \; & 100(1.06)^5 - 20(1.06)^4 + 100(1.06)^3 - 20(1.06)^2 - 20(1.06)^1 \\ = \; & £184.0026 \qquad (184.00 \text{ rounded}). \end{aligned}$$

2. The accumulation to $t = 10$ is the accumulation to $t = 5$ rolled up to time 10, i.e.

$$184.0026 \times V(10,5) = 184.0026 \times 1.06^5 = £246.2370 \qquad (246.24 \text{ rounded}).$$

3. The present value at $t = 0$ is the total accumulation to $t = 5$ rolled back to time 0, i.e.

$$184.0026 \times V(0,5) = 184.0026 \times 1.06^{-5} = 137.4974 \qquad (137.50 \text{ rounded}).$$

Solution 2

1. The accumulation to $t = 5$ is:

$$\begin{aligned} & 1000V(5,0) - 500V(5,1.5) + 100V(5,2) - 500V(5,2.5) - 500V(5,4) \\ = \; & 1000(1.08)^5 - 500(1.08)^{3.5} + 100(1.08)^3 - 500(1.08)^{2.5} - 500(1.08)^1 \\ = \; & -£205.3455 \qquad (-205.35 \text{ rounded}). \end{aligned}$$

2. The present value at $t = 0$ is the accumulation to $t = 5$ rolled back to time 0, i.e.

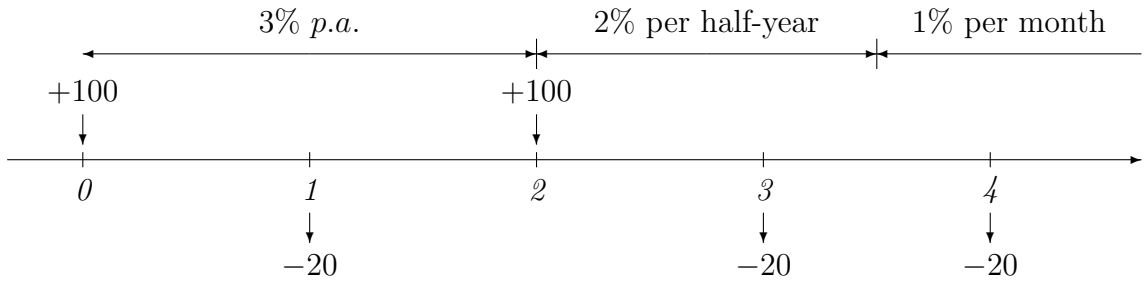
$$-205.3455 \times V(0,5) = -205.3455 \times 1.08^{-5} = -£139.7547 \qquad (-139.75 \text{ rounded}).$$

3. The value at $t = 2$ is the accumulation to $t = 5$ rolled back to time 2, i.e.

$$-205.3455 \times V(2,5) = -205.3455 \times 1.08^{-3} = -£163.0099 \qquad (-163.01 \text{ rounded})$$

Alternatively, show how can you obtain the value instead from the value at $t = 0$.

Solution 3 For convenience, we define the time unit to be 1 year, with 1/1/12 = 0, 1/1/13 = 1 and so on. We also add the interest rates to the diagram.



1. The present value on 01/01/12 ($t = 0$) is:

$$\begin{aligned}
&100V(0,0) - 20V(0,1) + 100V(0,2) - 200V(0,3) - 20V(0,4)\\
=&100 \times (1.03)^0 - 20 \times (1.03)^{-1} + 100 \times (1.03)^{-2} - 20 \times (1.03)^{-2} \times (1.02)^{-2}\\
&- 20 \times (1.03)^{-2} \times (1.02)^{-3} \times (1.01)^{-6}\\
=&£139.9872 \qquad (139.99 \text{ rounded}).
\end{aligned}$$

2. The accumulation on 01/05/17 ($t = 5\frac{4}{12}$) is the present value at $t = 0$ rolled up to time $5\frac{4}{12}$, i.e.

$$139.9872 \times V(5\frac{4}{12}, 0) = 139.9872 \times 1.03^2 \times 1.02^3 \times 1.01^{22} = £196.1704 \text{ (196.17 rounded)}.$$

The value on 01/07/15 ($t = 3\frac{6}{12}$) is the present value at $t = 0$ rolled up to time $3\frac{6}{12}$, i.e.

$$139.9872 \times V(3\frac{6}{12}, 0) = 139.9872 \times 1.03^2 \times 1.02^3 = £157.6026 \text{ (157.60 rounded)}.$$

Alternatively, show how can you obtain the value instead from the value at $t = 5\frac{4}{12}$.

Solution 4 If £X is paid into the bank account at $t = 0$, and all the later payments are made from the bank account, the accumulation at $t = 4$ when the last withdrawal is made will be:

$$\begin{aligned}
& X\ V(4,0) - 100\ V(4,1) - 100\ V(4,2) - 100\ V(4,3) - 100\ V(4,4)\\
=\ & X\ (1.055)^4 - 100\ (1.055)^3 - 100\ (1.055)^2 - 100\ (1.055)^1 - 100\ (1.055)^0\\
=\ & X\ (1.055)^4 - 434.226637.
\end{aligned}$$

If the payments are met exactly, this accumulation will be zero. Therefore, $X = £350.5150$ (350.51 rounded).

Solution 5

1. (a) The accumulation to the end of the payment stream of an annuity of 1 unit per time unit, payable in arrears for 10 time units, at an effective rate of interest of i per time unit.

   (b) The accumulation to the end of the payment stream of an annuity of 1 unit per time unit, payable in advance for 25 time units, at an effective rate of interest of i per time unit.

   (c) The present value at the start of the payment stream of an annuity of 1 unit per time unit, payable in arrears for 10 time units, at an effective rate of interest of i per time unit.

   (d) The present value at the start of the payment stream of an annuity of 1 unit per time unit, payable in advance for 20 time units, at an effective rate of interest of i per time unit.

2. To 5 decimal places,

   (a) $\dfrac{1.07^{10} - 1}{0.07} = 13.81645.$

(b) $1.07 \times \dfrac{1.07^{25}-1}{0.07} = 67.67647.$

(c) $\dfrac{1-1.07^{-10}}{0.07} = 7.02358.$

(d) $1.07 \times \dfrac{1-1.07^{-20}}{0.07} = 11.33560.$

Solution 6

1.
$$s_{\overline{n}|} = \sum_{r=1}^{n}(1+i)^{n-r} = (1+i)^n \sum_{r=1}^{n}(1+i)^{-r} = (1+i)^n a_{\overline{n}|}$$

2.
$$s_{\overline{n}|} = \sum_{r=1}^{n}(1+i)^{n-r} = 1 + \sum_{r=1}^{n-1}(1+i)^{n-r} = 1 + \ddot{s}_{\overline{n-1}|}.$$

3.
$$\ddot{a}_{\overline{n}|} = \sum_{r=0}^{n-1}(1+i)^{-r} = (1+i)^{-n} \sum_{r=1}^{n}(1+i)^{n-r-1} = v^n \ddot{s}_{\overline{n}|}$$

4.
$$\ddot{a}_{\overline{n}|} = \sum_{r=0}^{n-1}(1+i)^{-r} = 1 + \sum_{r=1}^{n-1}(1+i)^{-r} = 1 + a_{\overline{n-1}|}.$$